

**TELECOM USER ROLE MANAGEMENT SYSTEM
DATABASE ADMINISTRATION & SQL PRACTICAL ASSIGNMENT
COMPLETE SQL FORMAT SOLUTION**

-- SECTION A — DDL (DATA DEFINITION LANGUAGE)

-- =====

-- -----

-- QUESTION 1

-- Create database telecom_user_mgmt

-- -----

CREATE DATABASE telecom_user_mgmt;

USE telecom_user_mgmt;

-- -----

-- QUESTION 2

-- Create users table

-- -----

```
CREATE TABLE users (  
    user_id INT AUTO_INCREMENT PRIMARY KEY,  
    username VARCHAR(50) UNIQUE NOT NULL,  
    email VARCHAR(100) UNIQUE,  
    department VARCHAR(50),  
    status VARCHAR(20) DEFAULT 'ACTIVE',  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

-- -----

-- QUESTION 3

-- Create roles table

-- -----

```
CREATE TABLE roles (  
    role_id INT AUTO_INCREMENT PRIMARY KEY,  
    role_name VARCHAR(50) UNIQUE,  
    description VARCHAR(255)  
);
```

-- -----

-- QUESTION 4

-- Create permissions table

-- -----

```
CREATE TABLE permissions (  
    permission_id INT AUTO_INCREMENT PRIMARY KEY,  
    permission_name VARCHAR(100),  
    module_name VARCHAR(100)  
);
```

-- -----

-- QUESTION 5

-- Create user_roles table

-- -----

```
CREATE TABLE user_roles (  
    user_role_id INT AUTO_INCREMENT PRIMARY KEY,
```

```
user_id INT,
role_id INT,

FOREIGN KEY (user_id)
REFERENCES users(user_id),

FOREIGN KEY (role_id)
REFERENCES roles(role_id)
);

-----
-- QUESTION 6
-- Create audit_logs table
-----

CREATE TABLE audit_logs (
  log_id INT AUTO_INCREMENT PRIMARY KEY,
  user_id INT,
  action_performed VARCHAR(255),
  log_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

  FOREIGN KEY (user_id)
  REFERENCES users(user_id)
);

=====
-- SECTION B — DML (DATA MANIPULATION LANGUAGE)
=====

-----
-- QUESTION 7
-- Insert users
-----

INSERT INTO users (username, email, department)
VALUES
('aung_admin', 'aung@gmail.com', 'IT'),
('susu_billing', 'susu@gmail.com', 'Billing'),
('kyaw_support', 'kyaw@gmail.com', 'Customer Support'),
('mya_network', 'mya@gmail.com', 'Network Operations'),
('koko_manager', 'koko@gmail.com', 'Management');

-----
-- QUESTION 8
-- Insert roles
-----

INSERT INTO roles (role_name, description)
VALUES
('Admin', 'Full access to system'),
('Billing Officer', 'Manage billing operations'),
('Support Agent', 'Customer support access'),
('Network Engineer', 'Manage network operations'),
('Viewer', 'Read-only access');

-----
-- QUESTION 9
-- Insert permissions
-----
```

```

INSERT INTO permissions (permission_name, module_name)
VALUES
('Create User', 'User Management'),
('Update Billing', 'Billing'),
('View Reports', 'Reports'),
('Manage Network', 'Network'),
('Delete User', 'User Management');

```

```

-----
-- QUESTION 10
-- Assign roles to users
-----

```

```

INSERT INTO user_roles (user_id, role_id)
VALUES
(1, 1),
(2, 2),
(3, 3),
(4, 4),
(5, 5);

```

```

-----
-- QUESTION 11
-- Update one user status
-----

```

```

UPDATE users
SET status = 'INACTIVE'
WHERE user_id = 3;

```

```

-----
-- QUESTION 12
-- Delete one permission
-----

```

```

DELETE FROM permissions
WHERE permission_id = 5;

```

168 -- QUESTION 13
 169 -- Display all users
 170 -----
 171
 172 • SELECT * FROM users;

user_id	username	email	department	status	created_at
1	aung_admin	aung@gmail.com	IT	ACTIVE	2026-05-09 20:15:13
2	susu_billing	susu@gmail.com	Billing	ACTIVE	2026-05-09 20:15:13
3	kyaw_support	kyaw@gmail.com	Customer Support	INACTIVE	2026-05-09 20:15:13
4	mya_network	mya@gmail.com	Network Operations	ACTIVE	2026-05-09 20:15:13
5	koko_manager	koko@gmail.com	Management	ACTIVE	2026-05-09 20:15:13

175 -- QUESTION 14
 176 -- Display active users
 177 -----
 178
 179 • SELECT *
 180 FROM users
 181 WHERE status = 'ACTIVE';
 182

user_id	username	email	department	status	created_at
1	aung_admin	aung@gmail.com	IT	ACTIVE	2026-05-09 20:15:13
2	susu_billing	susu@gmail.com	Billing	ACTIVE	2026-05-09 20:15:13
4	mya_network	mya@gmail.com	Network Operations	ACTIVE	2026-05-09 20:15:13
5	koko_manager	koko@gmail.com	Management	ACTIVE	2026-05-09 20:15:13

184 -- QUESTION 15
185 -- Display users with roles
186 -----
187
188 • SELECT
189 u.username,
190 u.department,
191 r.role_name
192 FROM users u
193 JOIN user_roles ur
194 ON u.user_id = ur.user_id
195 JOIN roles r
196 ON ur.role_id = r.role_id;
197

username	department	role_name
aung_admin	IT	Admin
susu_billing	Billing	Billing Officer
kyaw_support	Customer Support	Support Agent
mya_network	Network Operations	Network Engineer
koko_manager	Management	Viewer

199 -- QUESTION 16
200 -- Count users by department
201 -----
202
203 • SELECT
204 department,
205 COUNT(*) AS total_users
206 FROM users
207 GROUP BY department;
208
209 -----

department	total_users
IT	1
Billing	1
Customer Support	1
Network Operations	1
Management	1

210 -- QUESTION 17
211 -- Display Billing department users
212 -----
213
214 • SELECT *
215 FROM users
216 WHERE department = 'Billing';
217
218 -----

user_id	username	email	department	status	created_at
2	susu_billing	susu@gmail.com	Billing	ACTIVE	2026-05-09 20:15:13

219 -- QUESTION 18
220 -- Display all roles and permissions
221 -----
222
223 • SELECT
224 r.role_name,
225 p.permission_name,
226 p.module_name
227 FROM roles r
228 CROSS JOIN permissions p;
229
230 -----

role_name	permission_name	module_name
Admin	Manage Network	Network
Admin	View Reports	Reports
Admin	Update Billing	Billing
Admin	Create User	User Management
Billing Officer	Manage Network	Network
Billing Officer	View Reports	Reports

231 -- QUESTION 19
 232 -- Sort users by username
 233 -----
 234
 235 • SELECT *
 236 FROM users
 237 ORDER BY username ASC;
 238
 239 -----

Result Grid

user_id	username	email	department	status	created_at
1	aung_admin	aung@gmail.com	IT	ACTIVE	2026-05-09 20:15:13
5	koko_manager	koko@gmail.com	Management	ACTIVE	2026-05-09 20:15:13
3	kyaw_support	kyaw@gmail.com	Customer Support	INACTIVE	2026-05-09 20:15:13
4	mya_network	mya@gmail.com	Network Operations	ACTIVE	2026-05-09 20:15:13
2	susu_billing	susu@gmail.com	Billing	ACTIVE	2026-05-09 20:15:13

240 -- QUESTION 20
 241 -- Total users in system
 242 -----
 243
 244 • SELECT COUNT(*) AS total_users
 245 FROM users;
 246
 247 -----
 248 -- SECTION D - TCL / DTL

Result Grid

total_users
5

-- QUESTION 21
 -- Transaction with COMMIT

START TRANSACTION;

INSERT INTO audit_logs (user_id, action_performed)
 VALUES (1, 'Created new telecom account');

COMMIT;

-- QUESTION 22
 -- Transaction with ROLLBACK

START TRANSACTION;

UPDATE users
 SET status = 'INACTIVE'
 WHERE user_id = 2;

ROLLBACK;

276 • SELECT *
 277 FROM users
 278 WHERE user_id = 2;

Result Grid

user_id	username	email	department	status	created_at
2	susu_billing	susu@gmail.com	Billing	ACTIVE	2026-05-09 20:15:13

-- QUESTION 23
 -- SAVEPOINT example

START TRANSACTION;

UPDATE users

```
SET status = 'INACTIVE'  
WHERE user_id = 1;
```

```
SAVEPOINT sp1;
```

```
UPDATE users  
SET status = 'INACTIVE'  
WHERE user_id = 4;
```

```
ROLLBACK TO sp1;
```

```
COMMIT;
```

```
-----  
-- QUESTION 24  
-- COMMIT vs ROLLBACK  
-----
```

```
-- COMMIT Example  
START TRANSACTION;
```

```
UPDATE users  
SET department = 'Management'  
WHERE user_id = 5;
```

```
COMMIT;
```

```
-- ROLLBACK Example  
START TRANSACTION;
```

```
UPDATE users  
SET department = 'Billing'  
WHERE user_id = 5;
```

```
ROLLBACK;
```

```
-- QUESTION 25  
-- Create billing_user  
-----
```

```
CREATE USER 'billing_user'@'localhost'  
IDENTIFIED BY 'password123';
```

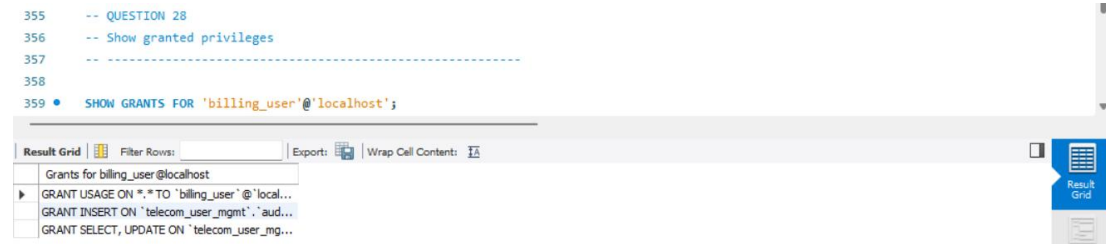
```
-----  
-- QUESTION 26  
-- Grant SELECT and UPDATE  
-----
```

```
GRANT SELECT, UPDATE  
ON telecom_user_mgmt.users  
TO 'billing_user'@'localhost';
```

```
-----  
-- QUESTION 27  
-- Grant INSERT on audit_logs  
-----
```

```
GRANT INSERT  
ON telecom_user_mgmt.audit_logs
```

TO 'billing_user'@'localhost';



-- QUESTION 29

-- Revoke UPDATE privilege

```
REVOKE UPDATE
ON telecom_user_mgmt.users
FROM 'billing_user'@'localhost';
```

-- QUESTION 30

-- Create viewer_user with read-only access

```
CREATE USER 'viewer_user'@'localhost'
IDENTIFIED BY 'viewer123';
```

```
GRANT SELECT
ON telecom_user_mgmt.*
TO 'viewer_user'@'localhost';
```

-- QUESTION 31

-- Only Admin users manage accounts

```
CREATE USER 'admin_user'@'localhost'
IDENTIFIED BY 'admin123';
```

```
GRANT ALL PRIVILEGES
ON telecom_user_mgmt.users
TO 'admin_user'@'localhost';
```

-- QUESTION 32

-- Billing department privileges

```
GRANT SELECT, UPDATE
ON telecom_user_mgmt.audit_logs
TO 'billing_user'@'localhost';
```

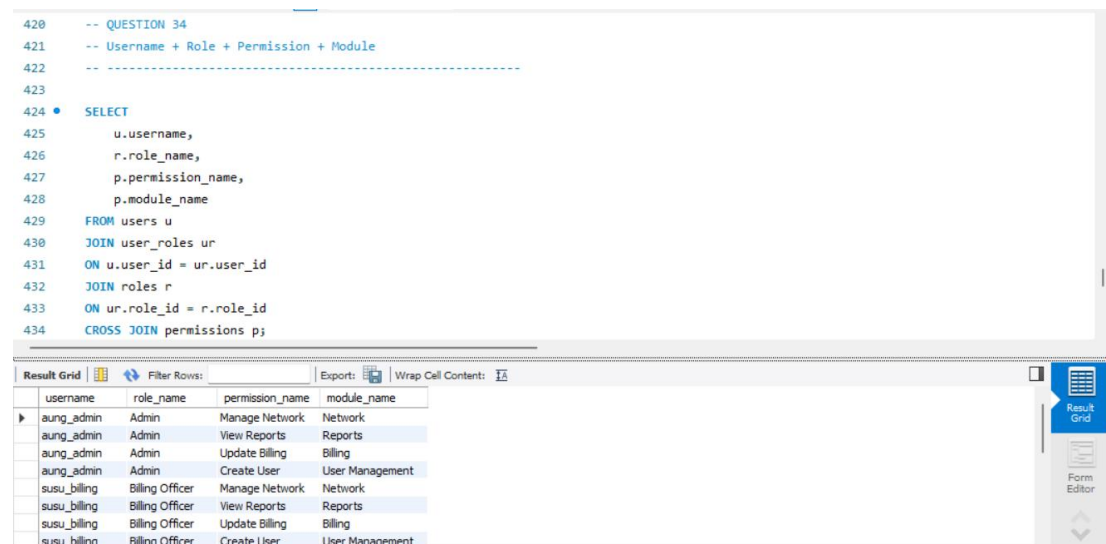
-- QUESTION 33

-- Support Agents read-only access

```
CREATE USER 'support_agent'@'localhost'
```

IDENTIFIED BY 'support123';

GRANT SELECT
ON telecom_user_mgmt.*
TO 'support_agent'@'localhost';



The screenshot shows a SQL IDE with a query editor and a results grid. The query is as follows:

```

420  -- QUESTION 34
421  -- Username + Role + Permission + Module
422  -- -----
423
424  SELECT
425      u.username,
426      r.role_name,
427      p.permission_name,
428      p.module_name
429  FROM users u
430  JOIN user_roles ur
431  ON u.user_id = ur.user_id
432  JOIN roles r
433  ON ur.role_id = r.role_id
434  CROSS JOIN permissions p;
  
```

The results grid displays the following data:

username	role_name	permission_name	module_name
aung_admin	Admin	Manage Network	Network
aung_admin	Admin	View Reports	Reports
aung_admin	Admin	Update Billing	Billing
aung_admin	Admin	Create User	User Management
susu_billing	Billing Officer	Manage Network	Network
susu_billing	Billing Officer	View Reports	Reports
susu_billing	Billing Officer	Update Billing	Billing
susu_billing	Billing Officer	Create User	User Management

QUESTION 35

RBAC Explanation in SQL Comment Format

ROLE-BASED ACCESS CONTROL (RBAC) IN TELECOM SYSTEMS

1. **SECURITY:** RBAC improves security by restricting unauthorized access to sensitive telecom data and operations.
2. **AUDITABILITY:** RBAC helps track user activities through audit logs, making monitoring and investigation easier.
3. **DATA PROTECTION:** Sensitive customer and billing information is protected from unauthorized viewing, editing, or deletion.
4. **ACCESS LIMITATION:** Users only receive permissions necessary for their roles. This minimizes accidental errors and insider threats.
5. **COMPLIANCE:** RBAC supports regulatory compliance and organizational security policies in telecom environments.